6TH GRADE MATH · UNIT 2: OPERATIONS OF FRACTIONS AND DECIMALS

Quilt Shop Makers

Teacher Guide & Answer Keys

Teacher

2–3 class periods

Project Overview

Students plan a class quilt. They operate with fractions and mixed numbers to measure and cut fabric and use decimal operations to price materials within a budget.

Division of fractions appears naturally ("how many pieces fit?"), and decimal multiplication appears in pricing.

Standards Addressed

- 6.NS.A.1 Divide fractions by fractions
- 6.NS.B.3 Add, subtract, multiply, and divide multi-digit decimals
- 5.NF / 6.NS review: add and subtract fractions and mixed numbers
- Honors: percent of waste and cost per unit comparisons

Pacing (2–3 class periods)

Day	Plan
Day 1	Launch with a picture of a quilt: how much fabric does it need? Part 1 fabric math.
Day 2	Part 2 cutting plan; model "how many $\frac{1}{8}$s fit?" with a ruler.
Day 3	Part 3 pricing; groups design and cost their quilt.

Materials

- Student handout with the Fabric Price Cards
- Rulers or yardsticks
- Colored pencils and grid paper
- Calculators for Part 3

How the Three Versions Differ

Level	What changes
CP2	Like-denominator steps, unit-fraction division, and simple decimal costs with starters.
CP1	Unlike denominators, inches-to-yards conversions, cutting plans with leftovers, and cost with change.
Honors	Eighth-yard rounding, waste percent, comparing fabric widths per square, and a budgeted design with profit.

Watch For (Common Misconceptions)

- Adding denominators when adding fractions.
- Flipping the wrong fraction when dividing.
- Lining up decimal points when multiplying decimals.

Extension Ideas

- Sew or glue a paper version of one quilt row.
- Find the total cost for a quilt twice as large in each direction.
- Research the price of a real quilt and compare.

Answer Key – CP2

Part 1: Measure the Fabric

#	Task	Answer
1	A strip is $\frac{3}{4}$ yard long. You cut off $\frac{1}{4}$ yard. How much is left?	$\frac{3}{4} - \frac{1}{4} = \frac{2}{4} = \frac{1}{2}$ yard
2	You have $1\frac{1}{2}$ yards of red and $\frac{3}{4}$ yard of yellow. How many yards is that in all?	$1\frac{1}{2} + \frac{3}{4} = 1\frac{2}{4} + \frac{3}{4} = 2\frac{1}{4}$ yards
3	Blue fabric comes in a $2\frac{1}{3}$ yard piece. You use $1\frac{2}{3}$ yards. How much is left?	$2\frac{1}{3} - 1\frac{2}{3} = \frac{7}{3} - \frac{5}{3} = \frac{2}{3}$ yard
4	Which is longer, $\frac{5}{8}$ yard or $\frac{2}{3}$ yard? Explain how you know.	$\frac{5}{8} = \frac{15}{24}$ and $\frac{2}{3} = \frac{16}{24}$, so $\frac{2}{3}$ yard is longer.

Part 2: Cut the Squares

#	Task	Answer
5	How many squares can you cut from $\frac{3}{4}$ yard of fabric?	$\frac{3}{4} \div \frac{1}{8} = \frac{3}{4} \times 8 = 6$ squares
6	The quilt has 20 squares. How many yards of fabric do you need?	$20 \times \frac{1}{8} = \frac{20}{8} = 2\frac{1}{2}$ yards
7	You want $\frac{1}{2}$ of the squares to be red. How many yards of red fabric is that?	$\frac{1}{2} \times 2\frac{1}{2} = 1\frac{1}{4}$ yards
8	A border strip is $2\frac{1}{2}$ yards long. You cut it into pieces that are $\frac{1}{2}$ yard long. How many pieces?	$2\frac{1}{2} \div \frac{1}{2} = 5$ pieces

Part 3: Price the Quilt

#	Task	Answer
9	Find the cost of 1.25 yards of red cotton.	1.25 × $7.60 = $9.50
10	Find the cost of 2.5 yards of batting and 2 spools of thread. Add to find the total.	2.5 × $4.50 = $11.25; 2 × $2.35 = $4.70; total $15.95

#	Task	Answer
11	Choose two more fabrics for your quilt and decide how many yards of each to buy. Find the total cost of everything.	Answers vary; check each product (yards × price per yard) and the sum.

Reflection

#	Prompt	Look for
R1	Why does dividing by $\frac{1}{8}$ give a bigger number?	Dividing by $\frac{1}{8}$ asks how many eighths fit, and there are 8 eighths in each whole, so the answer grows.
R2	Where do you line up decimals when you multiply money?	You do not line them up; multiply as whole numbers and count the decimal places in both factors.

Answer Key – CP1

Part 1: Fabric Math

#	Task	Answer
1	You need $1\frac{2}{3}$ yards of red and $2\frac{3}{4}$ yards of blue. How much fabric is that in all? What is the smallest amount you can buy if fabric is sold by the $\frac{1}{4}$ yard?	$1\frac{2}{3} + 2\frac{3}{4} = 1\frac{8}{12} + 2\frac{9}{12} = 4\frac{5}{12}$ yards. Buy $4\frac{1}{2}$ yards.
2	A $\frac{7}{8}$ yard strip is cut into a $\frac{1}{3}$ yard piece and the rest. How long is the rest, in yards and in inches?	$\frac{7}{8} - \frac{1}{3} = \frac{21}{24} - \frac{8}{24} = \frac{13}{24}$ yard = 19.5 inches
3	Write $\frac{5}{8}$ yard, 0.6 yard, and $\frac{2}{3}$ yard in order from least to greatest.	0.6 = $\frac{3}{5}$, $\frac{5}{8}$ = 0.625, $\frac{2}{3} \approx 0.667 \rightarrow 0.6 < \frac{5}{8} < \frac{2}{3}$
4	How many inches is $2\frac{3}{8}$ yards?	$2\frac{3}{8} \times 36 = \frac{19}{8} \times 36 = 85.5$ inches

Part 2: Cutting Plan

#	Task	Answer
5	How many $6\frac{1}{2}$-inch squares fit across a 44-inch width of fabric? How much is left over?	$44 \div 6\frac{1}{2} = 6$ R 5 → 6 squares with 5 inches left
6	A strip $6\frac{1}{2}$ inches tall gives 6 squares. How many strips (and how many yards of fabric) do you need for 20 squares?	20 ÷ 6 → 4 strips (24 squares). $4 \times 6\frac{1}{2} = 26$ inches = $\frac{26}{36} = \frac{13}{18}$ yard; buy $\frac{3}{4}$ yard.
7	The finished quilt is 4 squares across and 5 down. What are its finished length and width in feet? (12 inches = 1 foot)	$4 \times 6 = 24$ in = 2 ft wide; $5 \times 6 = 30$ in = $2\frac{1}{2}$ ft long
8	$\frac{2}{5}$ of the squares are blue and $\frac{1}{4}$ are yellow. How many squares is each? What fraction is left for other colors?	Blue $\frac{2}{5} \times 20 = 8$; yellow $\frac{1}{4} \times 20 = 5$; left $1 - \frac{2}{5} - \frac{1}{4} = \frac{7}{20}$ (7 squares)

Part 3: Cost & Change

#	Task	Answer
9	Find the cost of $\frac{3}{4}$ yard red, $\frac{3}{4}$ yard yellow, $1\frac{1}{4}$ yards blue, and 1 yard batting.	0.75(7.60) + 0.75(8.40) + 1.25(9.25) + 4.50 = 5.70 + 6.30 + 11.5625 + 4.50 = $28.06 (rounded)
10	Add 2 spools of thread. How much change is left from $60?	28.06 + 4.70 = $32.76; change $27.24
11	Design your own quilt (at least 3 colors). Find the yards to buy (in quarter yards) and the total cost, and show it is under $60.	Answers vary; check fractions of 20 squares, yardage rounded up to quarter yards, and the decimal cost total.

Reflection

#	Prompt	Look for
R1	Why do you round fabric amounts UP to the next quarter yard?	You cannot buy less than you need; rounding down leaves you short of fabric.
R2	Explain how you knew whether to multiply or divide in the cutting plan.	Dividing finds how many pieces fit in a length; multiplying finds the total length of a number of pieces.

Answer Key – Honors

Part 1: Yardage

#	Task	Answer
1	A pattern needs $1\frac{5}{6}$ yd of red, $2\frac{3}{8}$ yd of blue, and $\frac{11}{12}$ yd of yellow. Find the total and the least amount you must buy if fabric is sold by the $\frac{1}{8}$ yard (each color separately).	Total: $1\frac{20}{24} + 2\frac{9}{24} + \frac{22}{24} = 3\frac{51}{24} = 5\frac{1}{8}$ yd. Separately: red 1 $\frac{7}{8}$, blue $2\frac{3}{8}$, yellow 1 → $5\frac{1}{4}$ yd.
2	How much fabric (in inches) is wasted by rounding each color up?	Red $\frac{1}{24}$ yd = 1.5 in; blue 0; yellow $\frac{1}{12}$ yd = 3 in; total 4.5 in
3	A $\frac{5}{6}$ yd strip is cut into pieces of $\frac{5}{36}$ yd. How many pieces, and how many inches long is each?	$\frac{5}{6} \div \frac{5}{36}$ = 6 pieces; $\frac{5}{36}$ × 36 = 5 inches each
4	Is $\frac{7}{12}$ yd more or less than 0.58 yd? By how many inches?	$\frac{7}{12}$ ≈ 0.5833 yd, so it is more than 0.58 yd by about 0.0033 yd ≈ 0.12 inch.

Part 2: Efficient Cutting

#	Task	Answer
5	How many squares fit across each width, and what percent of the width is wasted on each?	Cotton: 6 squares, 5 in wasted → $\frac{5}{44}$ ≈ 11.4%. Denim: 58 ÷ 6.5 = 8 R 6 → 8 squares, 6 in → $\frac{6}{58}$ ≈ 10.3%.
6	For 30 squares, compare the inches of fabric length needed from cotton and from denim.	Cotton: 30 ÷ 6 = 5 strips × 6.5 = 32.5 in. Denim: 30 ÷ 8 → 4 strips × 6.5 = 26 in.
7	Using the Price Cards, which fabric costs less per square for 30 squares if you must buy in eighths of a yard?	Red cotton: 32.5 in is more than $\frac{7}{8}$ yd (31.5 in), so buy 1 yd = $7.60 → about $0.25 per square. Denim: 26 in fits in $\frac{3}{4}$ yd (27 in) = $6.94 → about $0.23 per square. Denim is cheaper per square.
8	A triangle piece is half of a square cut on the diagonal. How many triangles come from $\frac{3}{4}$ yard of 44-inch cotton?	27 in ÷ 6.5 = 4 strips × 6 squares = 24 squares × 2 = 48 triangles

Part 3: Budget Design

#	Task	Answer
9	What is the fixed cost of batting and thread? How much is left for fabric?	1.5 × 4.50 + 2 × 2.35 = 6.75 + 4.70 = $11.45; left $63.55
10	Design your quilt: give the fraction of squares in each fabric, the yardage to buy (in eighths), the cost of each, and the total.	Answers vary; check fractions sum to 1, yardage from strips, cost ≤ $75.
11	The auction sells the quilt for $3\frac{1}{2}$ times the material cost. What is the expected profit for your design?	Answers vary: profit = 2.5 × (material cost).

Reflection

#	Prompt	Look for
R1	When can rounding up fabric amounts cost more than the extra fabric is worth? How would you reduce waste?	When each color is rounded separately, waste adds up; combine pieces from one fabric or choose widths that divide evenly.
R2	Explain why a wider, more expensive fabric can be cheaper per square.	More squares fit across each strip, so fewer strips (less length) are needed, lowering the cost per square.